

१. **a. Discuss the planning approach for highway construction and development in Nepal.**  
**b. Discuss the controlling factors that should be considered in the alignment of highways.**  
**c. How are roads classified in Nepal according to the provisions made in NRS २०७०?**

Answer:

a. Nepal's rugged terrain, fragile geology, and limited financial resources requires highway planning with careful technical, environmental, and socio-economic considerations.

The planning approach for highway construction and development in Nepal are as follows:

**i. Traffic and engineering surveys:**

- Conduct traffic volume and Origin–Destination (O-D) surveys.
- Perform topographic, geological, geotechnical, and hydrological investigations.
- Collect environmental and socio-economic data.

**ii. Feasibility study:**

- Assess technical feasibility.
- Carry out economic and financial analyses (cost–benefit analysis).
- Evaluate environmental and social impacts.
- Estimate project costs and expected benefits.

**iii. Route selection:**

- Select the shortest, safest, and most economical alignment.
- Minimize earthwork, bridge construction, and land acquisition.
- Avoid landslide-prone areas, unstable slopes, flood plains, forests, and protected areas.
- Consider future road expansion.

**iv. Environmental and social assessment:**

- Conduct Initial Environmental Examination (IEE) or Environmental Impact Assessment (EIA) as required.
- Prepare environmental mitigation measures.
- Develop land acquisition, resettlement, and rehabilitation plans.
- Encourage public participation and stakeholder consultation.

**v. Geometric and structural design:**

- Design horizontal and vertical alignment.
- Determine pavement type and thickness.
- Design bridges, culverts, retaining walls, and drainage structures.
- Incorporate road safety features such as guardrails, signs, markings, and pedestrian facilities.

b. The selection of a highway alignment is governed by topographic, geological, economic, and social factors. A well-chosen alignment ensures safety, minimizes construction and maintenance costs, and reduces environmental impact.

The controlling factors that should be considered in highway alignment are listed below:

**I. Obligatory Points:**

Obligatory points are those control points governing the highway alignment that tells if the alignment must pass through or should be avoided because of engineering, economic, social, or environmental considerations.

- **Points to Pass Through:** The alignment may be deviated to connect intermediate towns, reach suitable bridge sites (where the river is straight and banks are stable), or cross mountain passes to attain necessary height.
- **Points to Avoid:** To minimize costs and conflicts, the alignment should avoid religious places, costly permanent structures, lakes/ponds, and marshy or water-logged lands.

## II. Topography:

- In mountainous regions, roads should follow natural ground contours as closely as possible to minimize massive excavation and filling.
- Design standards and speeds are determined by the cross-slope, categorized as Plain (0–10%), Rolling (10–25%), Mountainous (25–60%), or Steep (>60%).
- A primary goal is to achieve a balancing depth between cutting and filling to lower construction costs and reduce the need for external borrow pits.

## III. Geological stability:

- Alignments must strictly avoid active or old landslide zones, rockfall-prone areas, and weak geological formations such as faults or shear zones.
- Excessive cutting through hard solid rock should be avoided to control project time and expenses.

## IV. Geometric design:

- **Design Speed:** This is the most critical factor, as it determines curve radii, transition lengths, and required sight distances.
- **Gradients:** Alignments should maintain gentle gradients (desirably 5–7%). Where steep height must be gained in a short horizontal distance, hairpin bends or loops are introduced.
- **Sight Distance:** The alignment must ensure adequate Stopping Sight Distance (SSD) at all points, with Overtaking Sight Distance (OSD) provided where terrain allows.

c.

According to NRS 2070, roads in Nepal are broadly classified into administrative classification and technical classification.

### i. Administrative classification:

- It is important for assigning national importance and level of government responsible for overall management and methods of financing.
- According to this classification, roads are classified into:
  - a. **National highway:**
    - ✓ They are the main roads connecting East to west and North to South of the nation.
    - ✓ These serve for long distance travel, provide consistently higher level of service in terms of speed, travel and inter community mobility.
    - ✓ They are designate by 'NH' followed by two-digit number.
  - b. **Feeder roads:**
    - ✓ These roads serve by connecting district headquarters, major economic centers, tourism centers to national highways or other roads.
    - ✓ They are designated by letter F followed by 3-digit number.
  - c. **District roads:**
    - ✓ These are important roads within a district, serving areas of production and market and connecting with main highways.
  - d. **Urban roads:**
    - ✓ These roads serve within urban municipalities.

- ✓ According to National Urban Road Standard NURS,2076 urban roads are further classified into
  - **Arterial roads:**
    - These roads are meant for thorough traffic on continuous route.
    - Parking, loading and unloading activities are generally restricted and regulated.
    - Pedestrians are allowed to cross only at the intersections or at designated crossing.
  - **Sub-arterial roads:**
    - These roads are somewhat lower level of travel mobility than arterial roads.
    - Parking, loading and unloading are restricted and regulated.
    - Pedestrian are allowed to cross only at the intersections or at designated crossing.
  - **Collector roads:**
    - It is intended for collecting and distributing the traffic to and from local roads and also for providing access to sub-arterial or arterial roads.
    - They may be located in industrial, residential or business areas.
  - **Local roads:**
    - It provided access to residence, business or other property.
    - Such road doesn't carry large volume of traffic.

**ii. Technical/ Functional classification:**

- For assigning various geometric and technical parameters for design, roads are categorized into classes as follows:

**Class I:**

- ✓ They are highest standard roads with divided carriageway and access control with ADT 20,000 PCU or more in 20 years perspective period.
- ✓ Design speed adopted for design of this class roads in plain terrain is 120 kmph.

**Class II:**

- ✓ These are the roads with ADT of 5000-20000 PCU in 20 years perspective period.
- ✓ Design speed adopted for the design of this class of roads in plain terrain is 100 kmph.

**Class III:**

- ✓ They are the roads with ADT of 2000-5000 PCU in 20 years perspective period.
- ✓ Design speed adopted for design of this class of roads in plain terrain is 60 kmph.

**Class IV:**

- ✓ These are the roads with ADT of less than 2000 PCU in 20 years perspective period.
- ✓ Design speed adopted for design of this class of roads in plain terrain is 60 kmph.

The differences between NRS 2045 and NRS 2070 are:

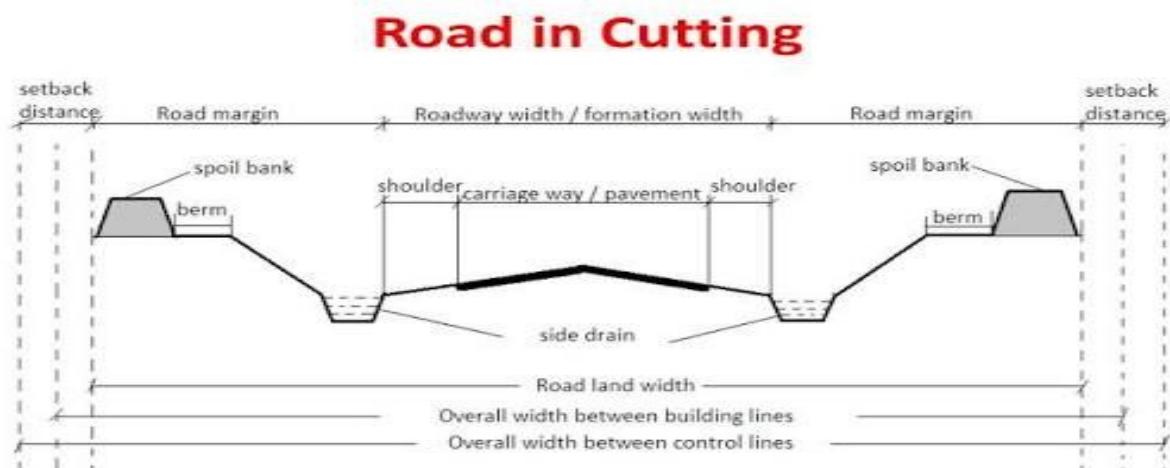
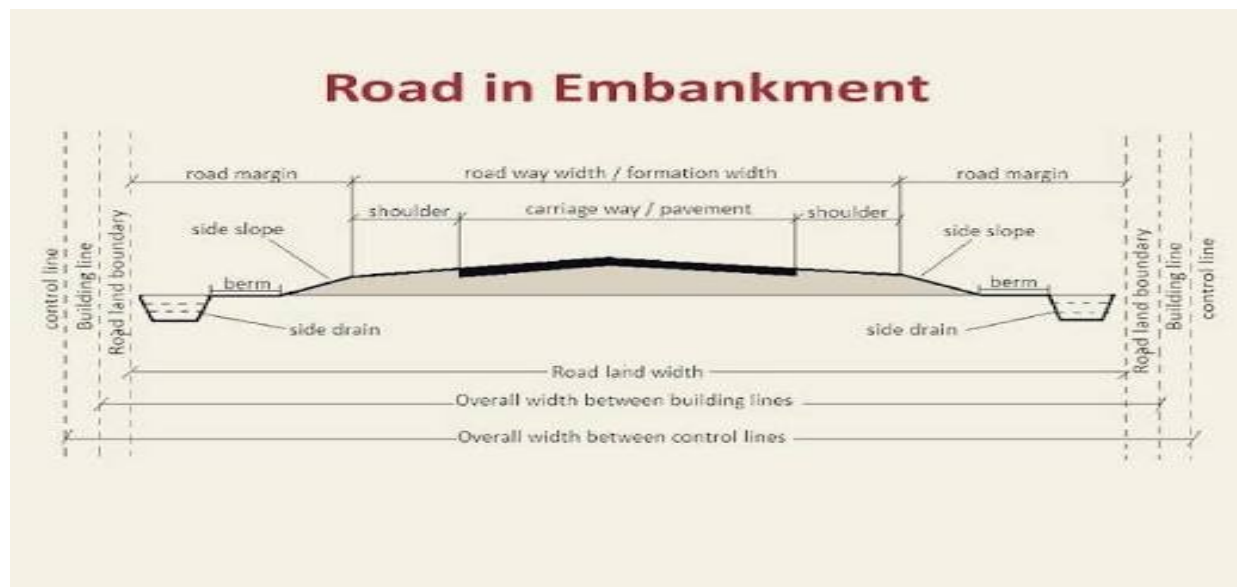
| NRS 2045                                                                                                                          | NRS 2070                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i. Administrative classes (National Highway, Feeder Road, District Road, Urban Road) present but with simpler functional grouping | i. Clear separation into administrative classes and technical/functional classes (Class I–IV) based on ADT and function, with more detailed criteria for each category. |
| ii. Uses ADT concept but with less detailed ranges.                                                                               | ii. Provides more explicit ADT ranges and links traffic volume to design standards (lanes, widths, sight distance, speed, pavement type)                                |

2. a. What are the elements of cross-section of a highway? Make the sketches of typical cross section for:  
 i. Embankment      ii. Cutting  
 b. What are the effects of centrifugal force acting on a vehicle moving on horizontal curve? How these effects are calculated?

Answer:

Highway cross-section is the section of road cut perpendicular to the centerline axis of the road. The elements of highway cross-section influence the life of the pavement as well as riding comfort and safety. It consists of many parts such as carriageway, shoulder, median strips, traffic separators, side drain, side slope, catch drains etc.

The elements of cross-section of a highway are listed below:



i. **Carriageway:**

- It is the strip of road meant for vehicular traffic movement also known as pavement width.
- Width of carriageway depends upon the width of traffic lanes and number of lanes.
- According to NRS 2070, The minimum lane width for a single lane road is 3.75 m.
- Two land road is provided with minimum of 3.5 m for each lane.

ii. **Shoulder:**

- They are the strips provided along the road edge and are intended for the accumulation of stopped vehicles.

- Shoulders serve as emergency lane for vehicles and also provide the lateral support for base and surface courses.
- According to NRS 2070, the width of shoulders on either side of carriageway shall be at least 0.75m.
- It is desirable that color and texture of shoulders be different from those of the carriageway.

**iii. Roadway width:**

- The sum of width of pavements or carriageway including separators and shoulders is called width of formation or roadway width.
- It is the top width of highway embankment or bottom width of highway cutting excluding the side drains.

**iv. Side slope:**

- According to NRS 2070, side slopes of embankment and cuttings depend upon the type of fill/ cut materials and height/depth of fill or cut.
- If natural cross slope of the ground is more than 1:5 then the ground should be cut with more than 2 m wide horizontal steps.

**v. Lay-bys, bus bays and bus stops:**

- Lay-bys are the intermittent shoulders sufficiently wide and long provided to meet the important function of shoulder where the continuous shoulder on either side cannot be provided from economic considerations.
- Bus-bays shouldn't be located too close near the intersections. It is desirable that they are located at least 40 m from the intersection on either side.
- Bus stops should be placed adjacent to the bus linear line of travel so that it doesn't need to pull over to left. It must be provided at the spacings of 200-00 m.

**vi. Right of way:**

- It is the width of land to be acquired for the road along its alignment. It must be adequate to accommodate all the cross-sectional elements of highway and reasonably provided for future development.
- According to NRS2070,
- ROW requirements for national highways= 50 m (25 m on either side of the road centerline)  
Feeder roads = 30 m (15 m on either side of road centerline)  
District roads= 20 m (10 m on either side of road centerline)

b.

A vehicle experiences an outward force called centrifugal force when a vehicle is traversing along a horizontal curve. This force acts away from the center of the curve and tends to push the vehicle outward. To counteract this force, superelevation (banking) and friction are provided.

The centrifugal force is calculated by the formula:

$$F = \frac{m v^2}{r}$$

Where :

$$F = \text{Centrifugal force (N)}$$

$$m = \text{Mass of the vehicle (kg)}$$

$$v = \text{Speed of the vehicle (m/s)}$$

$R = \text{Radius of the horizontal curve (m)}$

Centrifugal force causes skidding, overturning of vehicle, passenger discomfort, increased tyre and pavement wear.

The effects of centrifugal force are balanced by superelevation (e) and friction (f). the relationship is given by:

$$e + f = \frac{v^2}{127 R}$$

Where:

$e = \text{Super elevation (m / m)}$

$f = \text{Side friction factor}$

$v = \text{Speed (km / h)}$

$R = \text{Radius of the curve (m)}$

3. a. What are the special points to be considered in the selection of alignments of hill road? Discuss.  
b. Write short notes on:  
i. Hairpin bend  
ii. Prevention of landslides in hill roads.  
c. How to manage drainage in hill roads? Discuss.

Answer:

a. Hilly region of Nepal covers the mountainous & steep terrain having a cross slope of 25 to 60% and more. The road passing through the hilly terrain with a cross slope of 25% or more is generally termed as Hill Road. A hill road usually consists of either a river route or a ridge route. Road alignment in Nepal's hilly terrain is governed by a combination of **geological, topographic, drainage, economic, and geometric design factors**. The steep, unstable slopes and complex geology of the Himalayas and Middle Hills make the design and construction of hill roads more critical.



The special points to be considered in the selection of alignments of hill road are as follows:

i. **Geological Stability:**

Road alignment should pass through stable hill slopes and should avoid landslide and rockfall prone zones.

Construction activities might trigger landslide and rockfalls if the road alignment is to be passed through the area having unstable or weak hill slopes.

Excessive cutting works through hard solid rocks during construction must be avoided.

**ii. Cross drainage structures:**

Hilly region consists of large number of fast flowing, steep rivers that requires numerous cross-drainage works.

The road alignment should be selected such that the roadway consists of minimum number of cross drainage structures.

**iii. Geometric Design:**

Geometric design of hill roads must be done according to the design guidelines and standards for hill region.

The alignment should have minimum number of sharp curves and hairpin bends, gentler gradient.

**iv. Availability of construction materials:**

The road alignment should be selected in such a way that the construction materials are available near the construction site.

It reduces the transportation cost of materials and makes the construction more economical.

**v. Environment and weather:**

Rainfall (or Snowfall)  $\propto$  Altitude

As the altitude decreases, the number of cross-drainage works required increases.

At higher altitudes, the road pavement may witness snowfall during winter. This is why the alignment of hill roads should preferably be provided at an altitude between 900 m slopes exposed to sun.

**vi. Length:**

The cost of construction of a hill road per kilometer length is comparatively very high.

Hence, it should be ensured that the length of the road connecting two stations should be minimum possible.

**b. short notes:**

**i. Hairpin bends:**

The road passing through the hilly terrain with a cross slope of 25% or more is generally termed as Hill Road. A hill road usually consists of either a river route or a ridge route. The steep, unstable slopes and complex geology of the Himalayas and Middle Hills make the design and construction of hill roads more critical. Hill roads are also provided with hair pin bends and sharp curves wherever required.

Hairpin bends are those provided in hill road alignment introduced when it becomes necessary to attain height without covering substantial horizontal distance. They may be designed as either a circular curve with transition at each end or as a compound circular curve. In hairpin bends,

Minimum design speed= 20 kmph

Minimum radius of curvature = 15 m

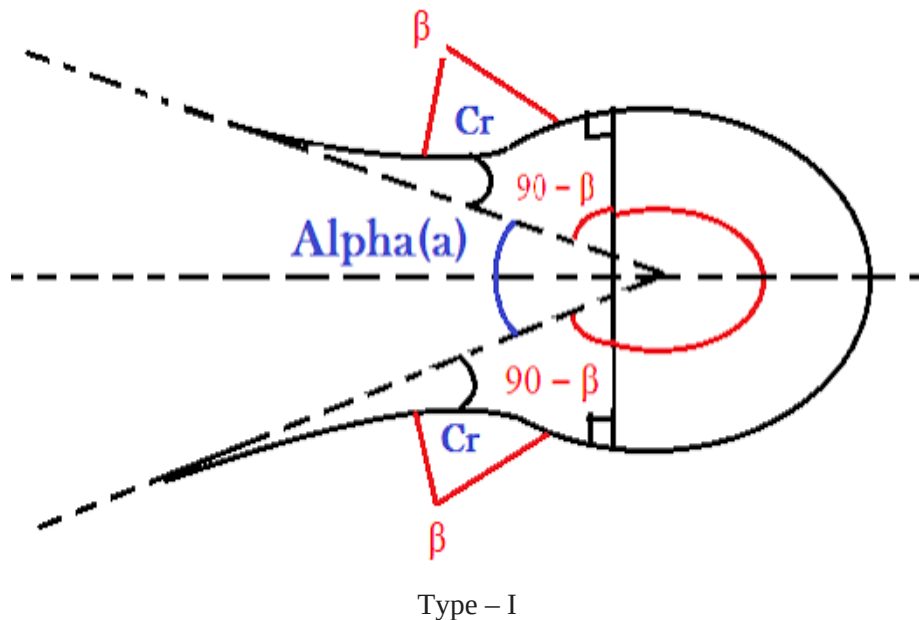
Minimum length of transition curve = 15 m

Maximum longitudinal gradient= 4%

Maximum superelevation =10%

At the hairpin bends, inner and outer edges of the roadway should be concentric with respect to the centerline of the pavement. A minimum intervening distance of 60 m should be provided between successive bends to enable driver to negotiate the alignment smoothly, where a number of hair pin bends have to be introduced. At hairpin bends preferably the full roadway width should be surfaced.

However, hairpin bends are not the desirable elements of hill road as the speed have to be substantially reduced. The construction cost also substantially increases because of the extensive volume of earthworks and provision of retaining structures.



## ii. Prevention of landslides in hill roads:

Landslide is the movement of a mass of soil, rock or debris down the slope under gravity. It is caused when gravitational pull exceeds the strength of soil. In the context of Nepal, rugged topography, unstable hills with large unstable slopes are commonly found in mountain and hilly regions. Few examples of the occurrence of landslide are Jure landslide, Sindhupalchowk, Bhirkot village landslide, Dolakha, Mugling-Naraynghad highway landslide disasters and so on.

Some of the measures for the prevention of landslides in hill roads are described below:

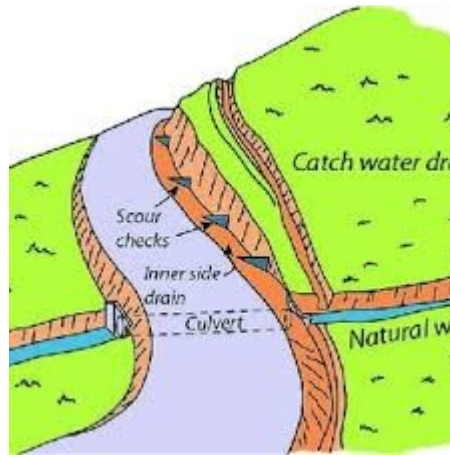
### ➤ Afforestation and vegetation:

Forests and vegetative systems strengthen the soil mass and prevent soil erosion. Hence, focusing more on afforestation and vegetation helps to solve the problem of landslide.



### ➤ Proper drainage provision:

Landslides get triggered due to heavy and prolonged rainfall. Rainwater washes the soil particles and reduces the shear strength of the soil. Hence, proper drainage infrastructures such as cascade, catch drains, side drains must be provided to intercept the surface runoff and prevent frequent landslides triggered by surface runoff.



➤ **Slope stabilization structures:**

The frequency of landslides can be effectively minimized by providing slope stabilization structures such as retaining walls, gabion walls, geosynthetics, soil nailing etc.



➤ **Planned land use:**

Land use must be planned before human settlement and construction of infrastructures. These constructions must be avoided near landslide-prone zones.

➤ **Controlled construction activities:**

Extreme care must be taken while carrying out construction activities. Sometimes, these construction activities itself trigger the unstable slope that results in occurrence of landslides.

c. Managing drainage in hill roads is essential for protecting the road structure, ensuring user safety, and preventing slope instabilities in geologically fragile terrain. Effective management requires an integrated approach combining surface drainage, subsurface drainage, cross-drainage structures, and bio-engineering measures.

i. **Surface drainage:**

Surface drainage involves the interception and removal of water from the road surface and the adjoining land.

- **Road Camber and Shoulders:** The carriageway and shoulders must be provided with a sufficient cross-slope (camber) to ensure surface water is effectively drained off to the side drains without allowing it to percolate into the subgrade.
- **Side Drains:** Constructed along the road edge (typically on the hill side), these channels collect runoff from the pavement and slopes. They must have sufficient capacity and longitudinal slope to carry away all collected water to the nearest outfall without causing erosion or cross-ruts.
- **Catch-water Drains (Intercepting Drains):** These are provided on the hill slope above the road cutting to intercept surface runoff from the upper reaches of the hill before it reaches the

road or weakens the cut slopes. The water is then safely diverted through cascades or chutes to natural gullies or culverts.

**ii. Subsurface drainage:**

Subsurface drainage is essential for removing water from below the ground surface to maintain the structural integrity of the road foundation.

- **Groundwater Table (GWT) Control:** A primary requirement for hill road design is to keep the highest level of the groundwater table preferably at least 1.2 metres below the subgrade level to prevent softening of the soil.
- **Control of Seepage:** If a seepage line (due to sloping impervious layers) reaches within 60–90 cm of the subgrade, it must be intercepted using longitudinal or transverse drainage trenches filled with filter material and perforated pipes.
- **Horizontal Drains:** In regions like Nepal where monsoon rains are intense, horizontal subsurface drains are often provided to divert internal slope water and reduce pore water pressure, which is a major cause of landslides.

**iii. Cross-drainage works:**

Cross-drainage structures are provided where the road alignment crosses natural streams, rivers, or drainage channels.

- **Culverts:** Used for small waterways up to 6 metres, such as pipe, box, slab, or arch culverts.
- **Bridges:** Required for linear waterways exceeding 6 metres.
- **Aqueducts:** These overhead structures can be used advantageously in hill roads to carry water across the road where traditional culverts are not feasible.
- **Causeways:** Used on less important roads to save construction costs, allowing water to pass over or under the road at specific points.

**iv. Bio-engineering:**

Bio-engineering uses vegetation to stabilize slopes and manage water movement.

- **Hydrological Control:** Vegetation intercepts rainfall and plant roots absorb soil moisture, which directly lowers pore water pressure and reduces the risk of slope failure.
- **Runoff Velocity Reduction:** Techniques such as live fascines (bundles of branches) and check dams made of bamboo or brushwood are built across gullies and slopes to slow down runoff, trap sediments, and prevent gully erosion.

**4. a. Give briefly how “traffic census” is conducted for estimating traffic volume for a given road.**

**b. How are road intersections planned? Enumerate the various traffic controls needed at an intersection.**

Answer:

a. A traffic census (traffic volume count) is the systematic counting and classification of vehicles passing a selected point on a road over a specified period. It is conducted to estimate the average traffic volume for planning, design, operation, and maintenance of highways.

Traffic census is conducted for estimating traffic volume for a given road as follows:

**i. Selection of Counting Location**

- Representative road section away from intersections, bus stops, and bottlenecks is chosen.
- The location reflecting the normal traffic conditions must be ensured.

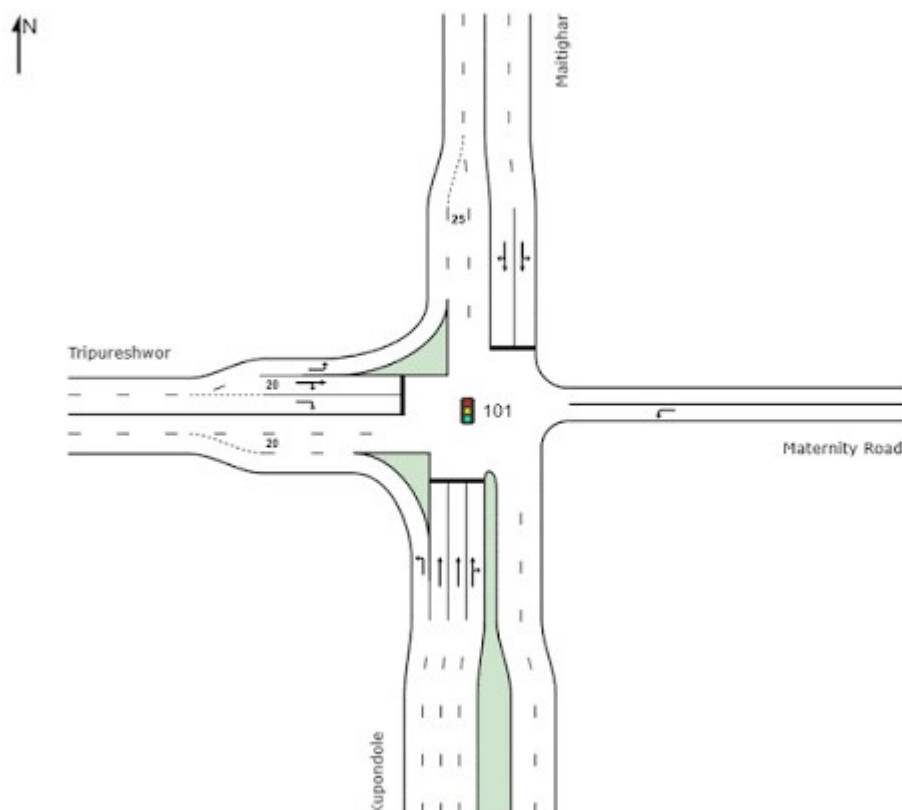
**ii. Selection of Counting Method**

- **Manual counting:** Enumerators record vehicles using tally sheets or hand counters.
- **Automatic counting:** Pneumatic tubes, inductive loops, video cameras, or electronic traffic counters are used.

**iii. Classification of Vehicles**

- Vehicles are grouped into categories such as:
  - i. Motorcycles
  - ii. Cars/jeeps/vans
  - iii. Buses
  - iv. Trucks
  - v. Tractors
  - vi. Non-motorized vehicles (where applicable)
- iv. **Counting Period**
  - Traffic is recorded at fixed intervals (e.g., every 15 minutes or hourly).
  - Counts may be conducted for 12 hours, 16 hours, or 24 hours, and on multiple weekdays and weekends to capture variations.
- v. **Recording Data**
  - Record the number and type of vehicles in each direction.
  - Note the date, time, weather conditions, and any unusual events affecting traffic.
- vi. **Data Analysis**
  - Calculate:
    - i. Average Daily Traffic (ADT)
    - ii. Annual Average Daily Traffic (AADT)
    - iii. Peak Hour Volume (PHV)
    - iv. Directional distribution
    - v. Traffic growth trends
- vii. **Preparation of Report**
  - The results are presented in tables, graphs, and charts.
  - These findings are used for highway planning, pavement design, capacity analysis, and maintenance planning.

b. Road intersection is defined as an area where two or more roads join or cross within which are included with the roadway and roadside facilities for traffic movement in that area. It is desirable that the intersection roads meet at or nearly at right angles.



Road intersections are planned by considering the following points:

- i. There should be adequate visibility for all the approaching vehicles on all converging roads.
- ii. Width of pavement at radius for turning should be sufficiently provided.
- iii. Area of conflict should be reduced to minimum by introducing properly designed traffic island and channelizing island.
- iv. Sudden change of path should be provided.

Various devices that are used to control, regulate and guide traffic are known as traffic control devices. These devices provide safe and efficient functioning of intersections by separating conflicting vehicle streams in time. There are different types of traffic control devices needed at the intersection. They are:

**i. Traffic sign:**

- Traffic sign conveys the specific information to driver in advance that allows him/her to be careful.
- It gives timely warning of hazardous situations and information of routes, directions and intersection in the highway.
- Traffic signs can be regulatory, warning signs or informatory signs.

**ii. Traffic signals:**

- Traffic signals are such control devices which alternatively direct the traffic to stop and process at level intersections.
- The main requirement of traffic signals is to draw attention, provide information and enough time to respond to the drivers.
- Traffic signals can be traffic control signals or pedestrian signal or special signal.

**iii. Traffic island:**

- Traffic island is the raised area provided at the level road to avoid or minimize the areas of major and minor conflicts.
- Island establishes physical channels through which vehicular traffic may be guided to travel and avoid conflict with other vehicular traffic.

**iv. Road markings:**

- It is used as means of controlling and guiding traffic.
- Traffic markings may be in the form of line pattern, words, symbols or reflectors on the pavement.
- Road markings promote smooth and harmonious flow of traffic along the guided paths of travel.

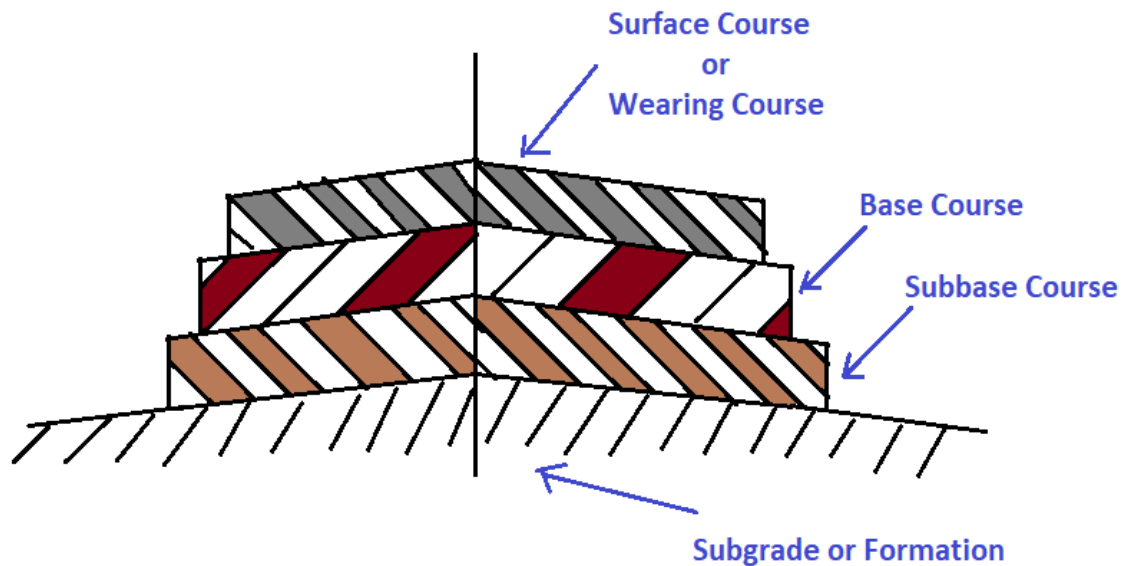
**5. a. What are the flexible and rigid pavements?**

**b. Explain briefly the construction procedures for a double bituminous surface dressing.**

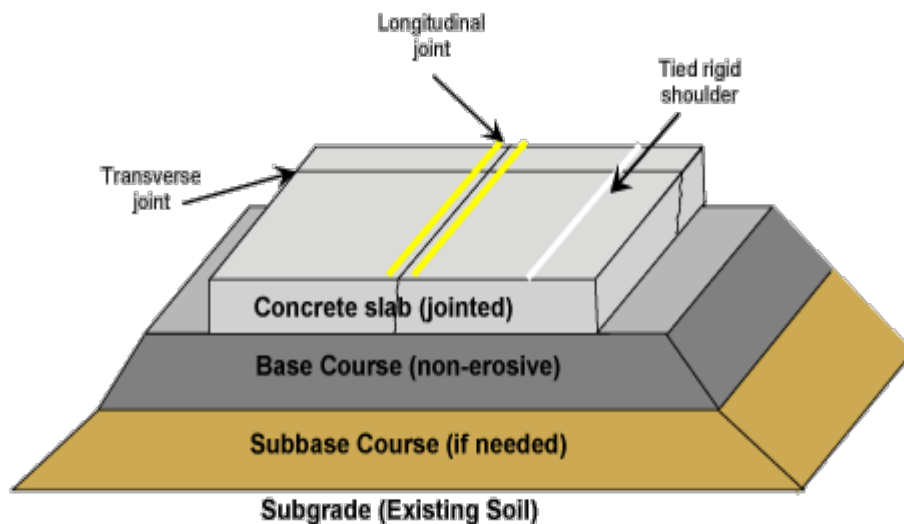
**c. What are the factors on which the thickness of pavement depends?**

Answer:

a. The pavements which have very low structural strength and are flexible in their structural behavior under the load are called flexible pavements. The flexible pavements layers reflect the deformation of the lower layers onto the surface of the layer by grain-to-grain contact. The strength of the pavement depends on the quality and thickness of each layer.



Rigid pavements are those which possess considerable flexural strength, made up of cement concrete which may either be plain, reinforced or prestressed. They have a slab action and are capable of transmitting the loads stresses through a wider area below.



b. In double bituminous surface dressing, immediately after laying the first coat, second application of binder is applied and followed by uniform spreading of the cover materials of smaller size aggregate and then rolled.

Construction procedure for double bituminous surface dressing:

- i. Preparation of existing surface:
  - Surface is prepared to proper profile before treatment, with proper rectification of ruts and depression.
  - All the dust particles or loose materials are removed.
  - Prime coat is applied if the existing base course has pervious surface.
- ii. Application of binder:
  - Uniform spraying of bituminous binder is done at the specified rate on a prepared surface using mechanical sprayer.
  - Excessive binder must not be applied at localized areas as it results in bleeding.
- iii. Application of stone chipping:
  - Cover material is spread as per the requirement to cover the surface uniformly.

- iv. Rolling of first coat:
  - Rolling is done with tandem roller (6 to 8 tonnes) starting from the edges proceeding towards the center longitudinally with overlapping not less than one third of the roller thread.
- v. Application of binder and stone chipping for second coat:
  - Immediately after the application of binder to the prepared surface, cover materials is spread as before.
  - Rolling is done in the safe manner.
- vi. Finishing and opening to traffic:
  - The surface is checked for longitudinal and cross profile using a straight edge of length 3 m and variation in surface greater than 6 mm are corrected.
  - The road section is opened to traffic after 24 hours.

c. The primary objective of selecting an appropriate thickness is to ensure the structure can carry anticipated traffic loads throughout its design life without excessive deformation or failure. The factors on which thickness of pavement depends are mentioned below:

i. Traffic loading:

Traffic is one of the most significant determinants of thickness. Higher traffic volumes and heavier axle loads directly necessitate thicker and stronger pavement layers. Specific considerations include:

- ✓ Magnitude and Frequency
- ✓ Axle Configuration
- ✓ Cumulative Standard Axles (CSA)

ii. Subgrade soil characteristics:

The subgrade is the natural soil foundation. A weak subgrade requires a thicker pavement structure to distribute the load effectively over a wider area. Key metrics used to assess this include:

- ✓ California Bearing Ratio (CBR)
- ✓ Modulus of Subgrade Reaction (k-value)
- ✓ Soil Properties

iii. Pavement materials:

The quality and properties of the materials used in the layers (aggregates, bitumen, or cement) affect the overall thickness required.

- ✓ **Flexible Pavements:** Strength depends on the quality and thickness of each individual layer (sub-base, base, binder, and surface courses), which transmit load through grain-to-grain contact.
- ✓ **Rigid Pavements:** The flexural strength of the cement concrete slab is paramount. If the slab is too thin, tensile stresses will exceed the concrete's strength, leading to premature cracking and failure.

iv. Environment and climatic factors:

Environmental factors influence the durability and required thickness of the pavement over time:

- ✓ **Rainfall and Drainage:** Poor drainage can soften the subgrade, requiring a thicker pavement to compensate for lost foundation strength.
- ✓ **Temperature Variations:** In rigid pavements, temperature changes cause warping and curling stresses, which must be accounted for in the slab thickness.

6. a. What do you understand by low-cost pavement? Explain application of otta seal.  
b. What do you mean by performance-based maintenance contract? Explain its elements.

Answer:

- A low-cost road is the road constructed at low cost and is capable of being maintained at low cost.
- It constructed using locally available materials, simple construction techniques, and minimum cost while providing an adequate level of service for roads carrying low to moderate traffic.
- Construction of low-cost road is preferred in developing countries like Nepal where large lengths of roads are to be constructed in the rural area with the limited available funds.
- It is commonly used for rural roads, village roads, and feeder roads, especially in developing countries like Nepal.
- Earthen roads, gravel roads and soil stabilized roads are the examples of such roads.

Otta seal is the bituminous surface dressing consisting of graded aggregates ranging from natural gravel to crushed rocks in combination with relatively soft binders with or without a sand cover seal. There are various types of otta seal in terms of number of layers, type of aggregate grading with or without sand cover seal. It is widely used for low-volume roads, especially in developing countries such as Nepal.

#### **Applications of Otta Seal:**

- i. **Rural and Feeder Roads**
    - Otta seal provides an economical all-weather surface for roads with low traffic.
  - ii. **Upgrading Gravel Roads**
    - It converts gravel roads into dust-free, durable roads without the high cost of asphalt concrete.
  - iii. **Low-Volume Roads**
    - It is suitable for roads carrying about **100–500 vehicles per day** (depending on design and local conditions).
  - iv. **Road Maintenance and Rehabilitation**
    - Otta seal is used to improve existing worn-out gravel or bituminous roads by providing a new wearing surface.
  - v. **Remote and Hilly Areas**
    - It is ideal where construction materials and heavy equipment are limited, making it suitable for Nepal's mountainous regions.
  - vi. **Dust Control**
    - It also helps to reduce the amount of dust generated from gravel roads, improving visibility, health, and driving comfort.
  - vii. **Areas with Limited Funds**
    - It is preferred where budget constraints make conventional asphalt pavements too expensive.
- b.
- A Performance-Based Maintenance Contract is a road maintenance contract in which the contractor is paid based on achieving specified performance standards, rather than on the quantity of work performed.
  - The contractor is responsible for maintaining the road in a predefined condition throughout the contract period.
  - Instead of paying for activities such as pothole patching or drain cleaning, the employer pays the contractor for ensuring that the road meets agreed performance levels.

#### **Elements of Performance Based Contract:**

- i. **Performance Standards**
  - It consists of clearly defined service levels that the road must meet.

- Examples:  
No potholes exceeding the specified size.  
Good pavement riding quality.  
Functional drainage system.  
Clear road signs and markings.

ii. **Scope of Work**

- The contract defines the maintenance activities to be performed, such as:  
Routine maintenance.  
Periodic maintenance.  
Emergency repairs.  
Bridge and drainage maintenance.  
Roadside vegetation control.

iii. **Performance Indicators**

- Performance based contract includes measurable criteria i.e., performance indicators used to assess the contractor's performance such as:  
Pavement condition.  
Surface roughness.  
Number and size of potholes.  
Shoulder condition.  
Response time to emergencies.

iv. **Monitoring and Inspection**

- Regular inspections by the client or supervising engineer.
- Performance is checked against the specified standards.

v. **Payment Mechanism**

- Payment is based on compliance with performance standards.
- Full payment is made if standards are met.
- Deductions or penalties are applied if standards are not achieved.

vi. **Risk Allocation**

- The contractor assumes greater responsibility for planning, maintenance methods, and achieving the required road condition.
- The employer focuses mainly on monitoring outcomes.

vii. **Contract Duration**

- Usually a multi-year contract (3–10 years) to encourage preventive maintenance and long-term performance.

7. **An aqueduct 3m wide and 2 m deep is used to carry water over a span of 7 m. Design a suitable section for the aqueduct. Make necessary assumption.**

Answer:

Given,

Width of aqueduct trough,  $B = 3.0 \text{ m}$

Water depth,  $h = 2.0 \text{ m}$

Span,  $L = 7.0 \text{ m}$

Since complete design data (grade of concrete, steel, loading details, freeboard, etc.) are not provided, reasonable assumptions are made for a preliminary design.

Let us assume:

RCC rectangular trough.

Concrete grade: M25.

Reinforcement steel: Fe500.

Unit weight of water =  $10 \text{ kN/m}^3$ .

Unit weight of RCC =  $25 \text{ kN/m}^3$ .

Freeboard =  $0.50 \text{ m}$ .

Simply supported over 7 m span.

Thickness of side walls = 250 mm.  
Bottom slab thickness = 300 mm.

Internal dimensions:

Width = 3.0 m

Water depth = 2.0 m

Freeboard = 0.5 m

Total inside depth = 2.5 m

External dimensions:

Side wall thickness = 0.25 m

Bottom slab thickness = 0.30 m

Therefore,

Overall width  
=  $3 + 0.25 + 0.25$   
= 3.50 m

Overall depth  
=  $2.5 + 0.30$   
= 2.80 m

Water pressure on bottom:  $= 10 \times 2 = 20 \text{ kN} / \text{m}^2$

Water carried:  $3 \times 2 \times 10 = 60 \text{ kN} / \text{m}$

Self-weight at the bottom:  $0.30 \times 3.5 \times 25 = 26.25 \text{ kN} / \text{m}$

Approximate self-weight of two side walls:  $2(0.25 \times 2.8 \times 25) = 35 \text{ kN} / \text{m}$

Total self-weight =  $26.25 + 35 = 61.25 \text{ kN} / \text{m}$

Total load:  $60 + 61.25 = 121.25 \text{ kN} / \text{m}$

Maximum Bending Moment:

For simply supported beam,

$$M = \frac{w L^2}{8} = \frac{121.25 \times 7^2}{8} = 742 \text{ kNm}$$

8. **a. Define principal planes and principal stresses. Write down the columb's failure theory.**  
**b. Describe the settlement of structure of soil. What is the difference between consolidation and compaction?**

Answer:

a. Principal planes are the planes at a point in a stressed body on which shear stress is zero and the normal stress is either maximum or minimum.

Principal stresses are the maximum and minimum normal stresses acting at a point on particular planes where the shear stress is zero.

They are given by:

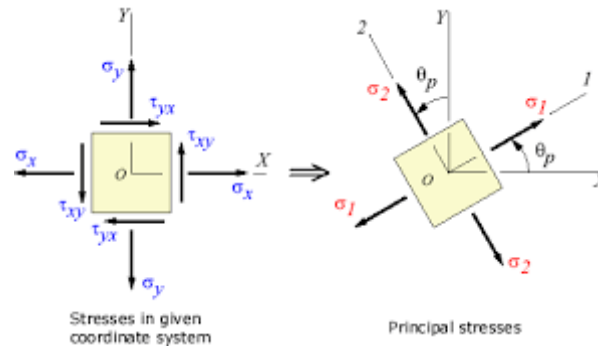
$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$\sigma_1$  = Major principal stress (maximum)

$\sigma_2$  = Intermediate principal stress

$\sigma_3$  = Minor principal stress (minimum)

For a two-dimensional stress system:



Mohr Column theory of shear strength is also known as Mohr-Columb's failure criteria. Initially, Mohr stated that "the shear strength of soil is the function of normal stress ( $\sigma$ ) acting on the failure plane.

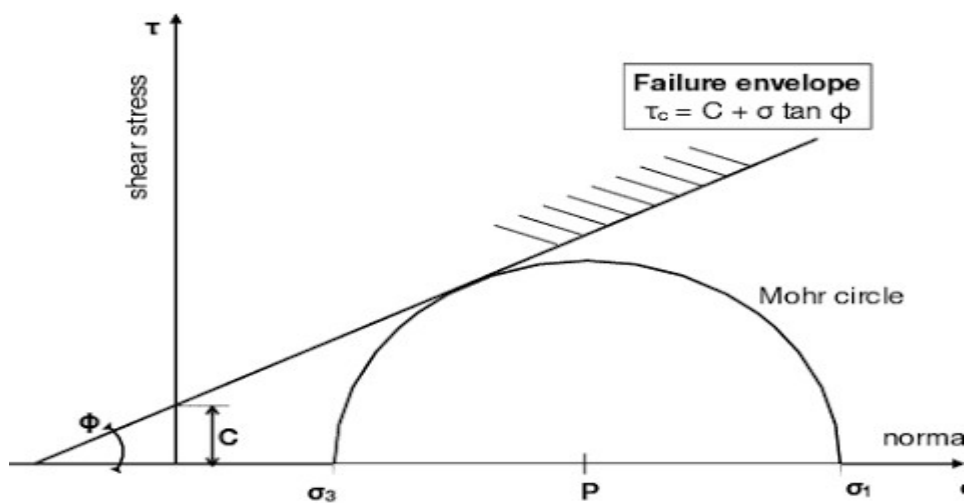
$$\text{i.e., } \tau = f(\sigma)$$

Later, Columb added on Mohr's theory stating "shear strength of soil is not only the function of normal stress ( $\sigma$ ) but also depends on surface particles forces between soil particles."

$$\text{i.e., } \tau = c + \sigma \tan \phi$$

This theory is known as Mohr- Columb's theory of shear strength.

Mohr-Coulomb failure criterion is represented by a **straight-line failure envelope** on the shear stress ( $\tau$ ) versus normal stress ( $\sigma$ ) graph.



Finally, Terzaghi modified Mohr-Columb's theory stating that "shear strength of soil is the function of effective stress not the total stress." This theory is known as modified Mohr-Columb's theory.

$$\text{i.e., } \tau = c' + \sigma' \tan \phi$$

b. The vertical downward movement (sinking) of a structure due to the compression or deformation of the soil beneath its foundation under applied loads is known as settlement. Excessive settlement can cause cracks, tilting, and structural damage. There are three types of settlement. They are:

- i. **Immediate (Elastic) Settlement**
  - Occurs immediately after the load is applied.
  - Caused by the elastic deformation of dry or partially saturated soils.
  - More common in sandy and gravel soils.
- ii. **Primary Consolidation Settlement**
  - Occurs mainly in saturated clay soils.
  - Caused by the expulsion of water from the soil pores under sustained loading.
  - Takes months or years to complete.
- iii. **Secondary Compression Settlement**
  - Occurs after primary consolidation is complete.
  - Caused by the gradual rearrangement of soil particles under constant load.
  - Significant in highly compressible organic soils and soft clays.

Soil compaction is the process of densification of soil by removing the air voids present in the soil particles under the action of mechanical energy. Similarly, consolidation is the process of compression of water voids due to which expulsion of water from the void occurs under the effect of applied load.

The differences between soil compaction and consolidation are given in points.

|    | Compaction                                                           |    | Consolidation                                                                           |
|----|----------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------|
| 1. | Densification of soil under the action of mechanical energy.         | 1. | Compression of water voids under the effect of sustained applied load.                  |
| 2. | Compaction is immediate process.                                     | 2. | Consolidation is slow and time-dependent process.                                       |
| 3. | It occurs due to reduction of air voids.                             | 3. | It occurs due to reduction of water filled voids.                                       |
| 4. | Compaction increases dry density and strength of soil.               | 4. | Consolidation causes settlement and increase in effective stress.                       |
| 5. | Compaction is generally artificial and intentional process.          | 5. | Consolidation is generally natural or load-induced.                                     |
| 6. | Equipment such as roller, rammer, vibrators are used for compaction. | 6. | No mechanical equipment is required for consolidation. It occurs under structural load. |

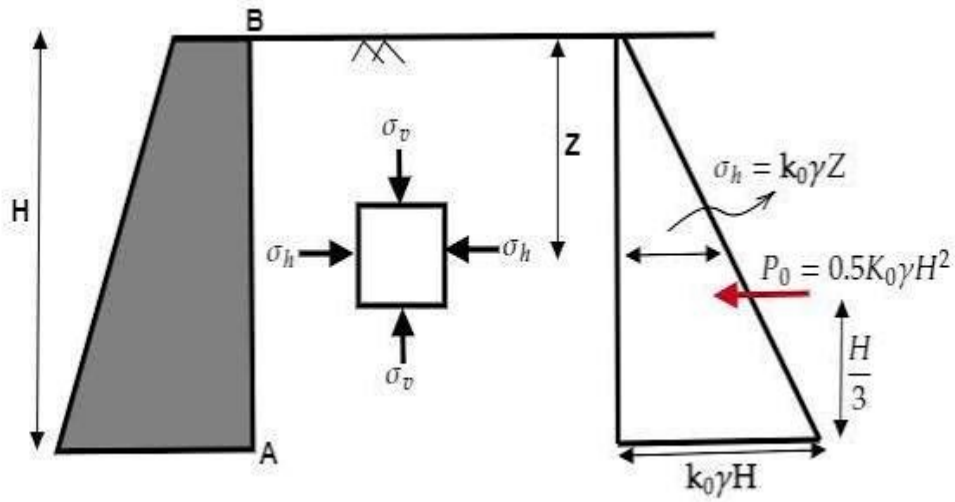
9. **What are the assumption of Rankine's theory? Derive the expression for active and passive pressures.**

Answer:

The expression for active earth pressure on the vertical back face of earthwork supporting granular soil having horizontal surface is given by Rankine's theory.

The assumptions for Rankine's theory:

- i. The back of the retaining wall is vertical and smooth (no wall friction).
- ii. The backfill is dry, homogeneous, cohesionless (granular) soil.
- iii. The ground surface is horizontal.
- iv. The soil is in the plastic equilibrium state.
- v. Failure occurs along a plane surface.



Let us consider a soil element at a depth  $h$  below the ground surface.

Vertical stress due to overburden:

$$\sigma_v = \gamma h$$

where:

$\gamma$  = unit weight of soil ( $\text{kN/m}^3$ )

$h$  = depth below ground level (m)

The horizontal stress under active condition is

$$\sigma_h = K_a \sigma_v$$

where  $K_a$  is the coefficient of active earth pressure.

According to Rankine's failure criterion,

$$K_a = \frac{1 - \sin \phi}{1 + \sin \phi}$$

or

$$K_a = \tan^2 \left( 45^\circ - \frac{\phi}{2} \right)$$

where

$\phi$  = angle of internal friction of soil.

Substituting

$$\sigma_h = K_a \gamma h$$

For a retaining wall of height  $H$ ,

Pressure at the top:

$$p=0$$

Pressure at the bottom:

$$p = K_a \gamma H$$

The total active earth pressure equals the area of the pressure triangle.

$$P_a = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$P_a = \frac{1}{2} (K_a \gamma H) (H)$$

Therefore,

$$P_a = \frac{1}{2} K_a \gamma H^2$$

This force acts horizontally on the wall.

Since the pressure distribution is triangular, the resultant acts at the centroid of the triangle.

Hence,

$$\text{Point of application} = \frac{H}{3} \text{ above the base}$$

10. **Discuss the factors determining the selection of particular type of foundation for a bridge. Illustrate your answer with examples.**

Answer:

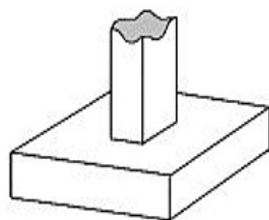
a. The selection of a specific type of foundation for a bridge is very essential that ensures the safe transfer of loads from the superstructure to stable soil or rock layers.

The primary factors determining the selection of a particular type of foundation for a bridge are listed below:

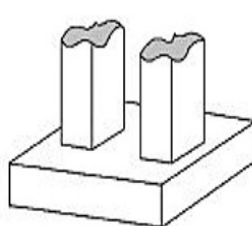
i. **Geological and geotechnical conditions:**

**Bearing Capacity:** If firm soil or sound rock with adequate bearing capacity is available near the ground surface, shallow foundations (such as isolated or strip footings) are preferred. If the upper soil layers are weak, compressible, or highly variable, deep foundations are necessary to reach stronger strata at greater depths.

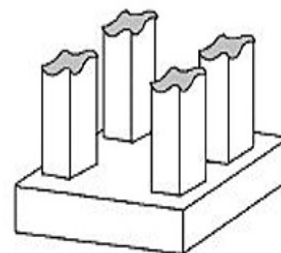
**For example:** Shallow foundation is used in a bridge in a hilly region where sound rock is found at a depth of 2 metres.



Isolated Footing



Continuous footing

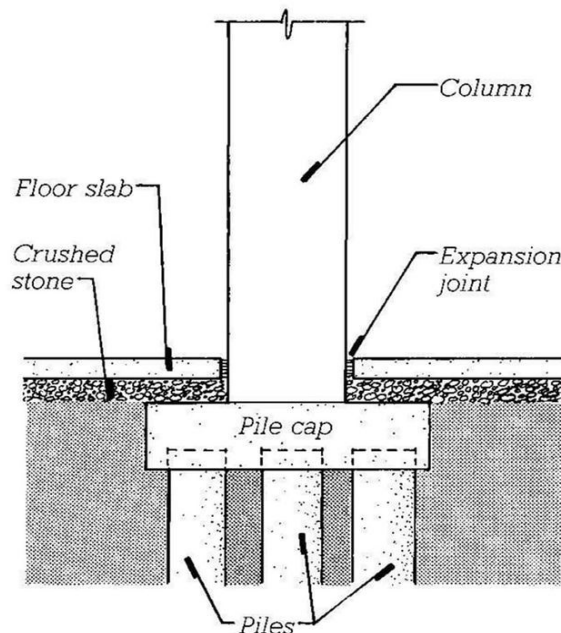


Mat footing

- ii. **Soil Type:** Soft clay or loose sand may necessitate pile foundations to transfer loads through skin friction and end bearing. In contrast, a site with sound rock near the surface allows for simpler open foundations.

**For example:** Pile foundation is used in a highway bridge constructed over a marshy wetland with deep layers of soft silt.

### Important Details about Piles



iii. **Hydrological factors:**

**Scour Resistance:** Foundations must be fixed below the calculated maximum scour depth to prevent the structure from being undermined during high-velocity flood events.

**Lateral Forces:** In fast-flowing rivers with heavy lateral forces, well foundations (caissons) are often selected because they provide superior resistance compared to piles.

iv. **Loads on the structures:**

**Heavier Superstructures:** Heavier structures, such as arch bridges, require very strong and rigid foundations to handle significant thrust and weight.

**Lightweight Structures:** Lighter bridges, like steel trusses or cable-supported crossings, may be feasible on slightly weaker strata where extensive deep foundations might be otherwise required for heavier concrete designs.

Well foundation is constructed for a bridge over a large, fast-flowing river like the Koshi.

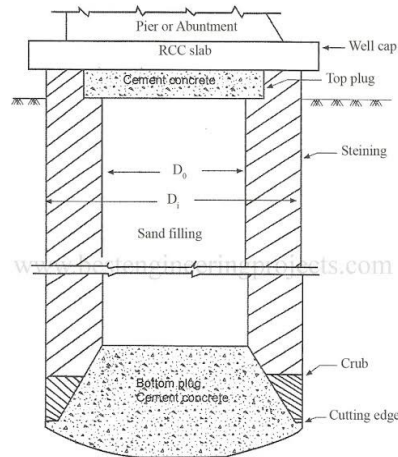


Fig 1 Parts of a Well Foundation

- v. In areas with restricted working space, pile foundations are often preferred because the equipment required is generally more compact than that needed for large-scale well sinking.

Mat foundation is constructed for a bridge abutment where the columns are very closely spaced or the soil has low uniform bearing capacity.

- vi. **Waterlogged Areas:** In marshy or waterlogged terrain, pile foundations are the standard choice as they can be driven or bored effectively in such conditions.
- vii. **Seismic Risks:** In earthquake-prone regions like Nepal, foundations must be designed for ductile detailing and stability against liquefaction.
- viii. **Economic factors:**

**Cost:** Pile foundations are economical for small to medium-scale bridges. While well foundations are expensive, they become the more economical choice for major river bridges where the depth of scour and load magnitude are extreme.

**Material Availability:** The proximity and cost of materials like cement, steel, and aggregates also influence the choice, as some foundation types require more specialized materials or fabrication.